

Mohamed Abuella

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Scholar: Mohamed Abuella

Summary

My research develops data-driven decision analytics for integrated energy systems to support the transition toward net-zero energy infrastructures. I combine artificial intelligence, optimization, and game-theoretic modeling to analyze the planning and operation of modern energy systems with high penetration of renewable resources and emerging technologies such as hydrogen.

My work spans energy forecasting, spatiotemporal analytics, and strategic infrastructure planning, with applications in renewable integration, energy efficiency, and sector coupling. Through interdisciplinary research that bridges power systems engineering, artificial intelligence, and policy-oriented modeling, I aim to develop analytical tools that support resilient, efficient, and sustainable energy transitions.

I am particularly interested in advancing AI-enabled decision support for complex socio-technical energy systems and collaborating across academia, industry, and policy stakeholders to accelerate the deployment of sustainable energy solutions.

Education

University of North Carolina at Charlotte (UNCC)

Ph.D in Electrical Engineering, GPA 4.0

USA

2014–2018

Southern Illinois University at Carbondale (SIUC)

M.Sc in Electrical and Computer Engineering, GPA 4.0

USA

2010–2012

Higher Polytechnic Institute & College of Industrial Technology at Misurata

DipHE in Instrumentation 82%, and B.Tech Electromechanical Engineering, 86%

Libya

2002–2008

Experience

Research Fellow

Northumbria University, Department of Mathematics, Physics and Engineering

UK
May 2024–Present

Supporting the delivery of the “Hydrogen Integration for Accelerated Energy Transitions (HI-ACT)” research project.

○ Acquired Expertise: *Integrated Energy Systems, Game Theory, Strategic & Policy Modeling, Hydrogen Integration, Energy Transition*

Researcher

Halmstad University

Sweden
April 2022–April 2024

Postdoctoral Researcher at the Center for Applied Intelligent Systems Research (CAISR). Research on AI for Sustainability applying Machine Learning techniques.

○ Acquired Expertise: *Integrated Industry-Academia Collaboration, Big-Data & Spatiotemporal Analysis, Explainable AI (XAI), Research Methodologies & Project Supervision.*

Lecturer

College of Industrial Technology at Misurata

Libya
February 2020–March 2022

Taught Electrical Circuits, Electrical Measurements, Math 101.

○ Acquired Expertise: *Curriculum Revision & Preparing, Dedication, Listening, "Try to Modeling the Student's Way of Thinking".*

Research Assistant

Energy Production and Infrastructure Center (EPIC) at UNC Charlotte

USA
2014–2019

Statistical and Predictive Analytics to Modernize the Grid and Optimize its Integration of Renewables, focusing on Solar Energy Resources.

○ Acquired Expertise: *Energy Analytics, Energy Markets, Renewable Energy & Supply Chain, Machine Learning.*

M.Sc Research

Department of Electrical and Computer Engineering at SIUC

USA
2010–2012

Optimization for Electric Power Systems Including Wind Power.

○ Acquired Expertise: *Power Systems Analysis, Operation and Planning, Systems Optimization, Smart Grid.*

Teaching Assistant and Lab Instructor

College of Industrial Technology at Misurata

Taught Mathematics, Power Systems Analysis, and PLC.

○ Acquired Expertise: *Teaching, Tutorials, Lab Modeling & Simulations.*

Libya

2008–2009

Electrical Technician

Residential Electrical Wiring and Water & Wastewater Company

Wiring and maintain electrical control equipment.

○ Acquired Expertise: *Electrical Wiring & Installations, Maintenance & Operation.*

Libya

2000–2008

Recognitions

Best Paper Award (Third Prize): RPG 2025 – 14th Int. Conf. on Renewable Power Generation, Shanghai, China 2025

Outstanding Reviewer: IEEE Transactions on Sustainable Energy 2017

Third Prize for Student Papers: The 47th North American Power Symposium 2015

The Institute Prize for team of Energy Analytics Lab at UNCC: Global Energy Forecasting Competition 2014

The 1st Place: Department of Electromechanical Engineering at College of Industrial Technology 2008

Publications

Wrote dozen of published papers, for the complete list of publications, please see my profile at Google Scholar, which is named as: [Mohamed Abuella](#). Accumulating over 700 citations and current H-index is **9**. my [ORCID](#).

Journal Articles

1. M. Abuella, A. Allahham, S. L. Walker, "Game Theory Approaches to Hydrogen Infrastructure Investment Planning in Great Britain: A Comparative Analysis of Competitive and Cooperative Frameworks," *International Journal of Hydrogen Energy*, vol. 220, p. 154064, 2026. DOI: 10.1016/j.ijhydene.2026.154064.
2. M. Abuella, A. Allahham, N. A. Rufa'I, S. L. Walker, "Hydrogen-Based Decarbonization Strategies for Residential Heating: An Energy Efficiency and Conservation Analysis in the North of Tyne Region," *Energies*, vol. 18, no. 23, 2025, Article 6237. doi: 10.3390/en18236237.
3. M. Abuella, H. Fanaee, S. Nowaczyk, S. Johansson, E. Faghani, "Time-Series Analysis Approach for Improving Energy Efficiency of Fixed-Route Passenger Vessel in Short-Sea Shipping," *Ocean Engineering*, 334, 121555, 2025.
4. M. Abuella, M. Atoui, S. Nowaczyk, S. Johansson, E. Faghani, "Spatial Clustering Approach for Vessel Path Identification," *IEEE Access*, 2024.
5. M. Abuella and B. Chowdhury, "Forecasting of Solar Power Ramp Events: A Post-Processing Approach," *Renewable Energy*, 133, 1380-1392, 2019.
6. M. Abuella and B. Chowdhury, "Improving Combined Solar Power Forecasts Using Estimated Ramp Rates: Data-Driven Post-Processing Approach," *IET Renewable Power Generation*, 12(10), 1127-1135, 2018.

Manuscript Under Review

M. Abuella, A. Allahham, S. L. Walker, "A Policy-Navigation Framework for Exploring Hydrogen Integration Pathways in Great Britain Toward Net Zero," Submitted to *Energy Policy*, February 28, 2026.

Conference Papers

1. M. Abuella, A. Allahham, S. L. Walker, "Socio-Technical, Electrification, and Hydrogen-Driven Pathways for Residential Heating Decarbonisation in the North of Tyne," in *9th International Conference on Environment Friendly Energies and Applications (EFEA 2025)*, 4–5 December 2025, Newcastle upon Tyne, UK.
2. M. Abuella, A. Allahham, S. L. Walker, "A Game-Theoretic Framework for Hydrogen Integration to Accelerate Energy Transitions in Great Britain," in *14th International Conference on Renewable Power Generation (RPG 2025)*, 24–26 October 2025, Shanghai, China.
3. M. Abuella, M. Atoui, S. Nowaczyk, S. Johansson, E. Faghani, "Data-Driven Explainable Artificial Intelligence for Energy Efficiency in Short-Sea Shipping," in *ECML PKDD*, 2023.
4. M. Abuella and B. Chowdhury, "Adjusting Post-Processing Approach for Very Short-Term Solar PV Power Forecasts," in *2021 IEEE MI-STA*, 2021.
5. M. Abuella and B. Chowdhury, "Qualifying Combined Solar Power Forecasts in Ramp Events' Perspective," in *IEEE Power and Energy Society General Meeting*, 2018.
6. M. Abuella and B. Chowdhury, "Forecasting Solar Power Ramp Events Using Machine Learning Classification Techniques,"

in *IEEE PEDG*, 2018.

7. M. Abuella and B. Chowdhury, "Hourly Probabilistic Forecasting of Solar Power," in *North American Power Symposium (NAPS)*, 2017.
8. M. Abuella and B. Chowdhury, "Random Forest Ensemble of Support Vector Regression Models for Solar Power Forecasting," in *IEEE ISGT*, 2017.
9. M. Abuella and B. Chowdhury, "Solar Power Forecasting Using Support Vector Regression," in *ASEM International Annual Conference*, 2016.
10. M. Abuella and B. Chowdhury, "Solar Power Forecasting Using Artificial Neural Networks," in *North American Power Symposium (NAPS)*, 2015.
11. M. Abuella and B. Chowdhury, "Solar Power Probabilistic Forecasting by Using Multiple Linear Regression Analysis," in *IEEE SoutheastCon*, 2015.